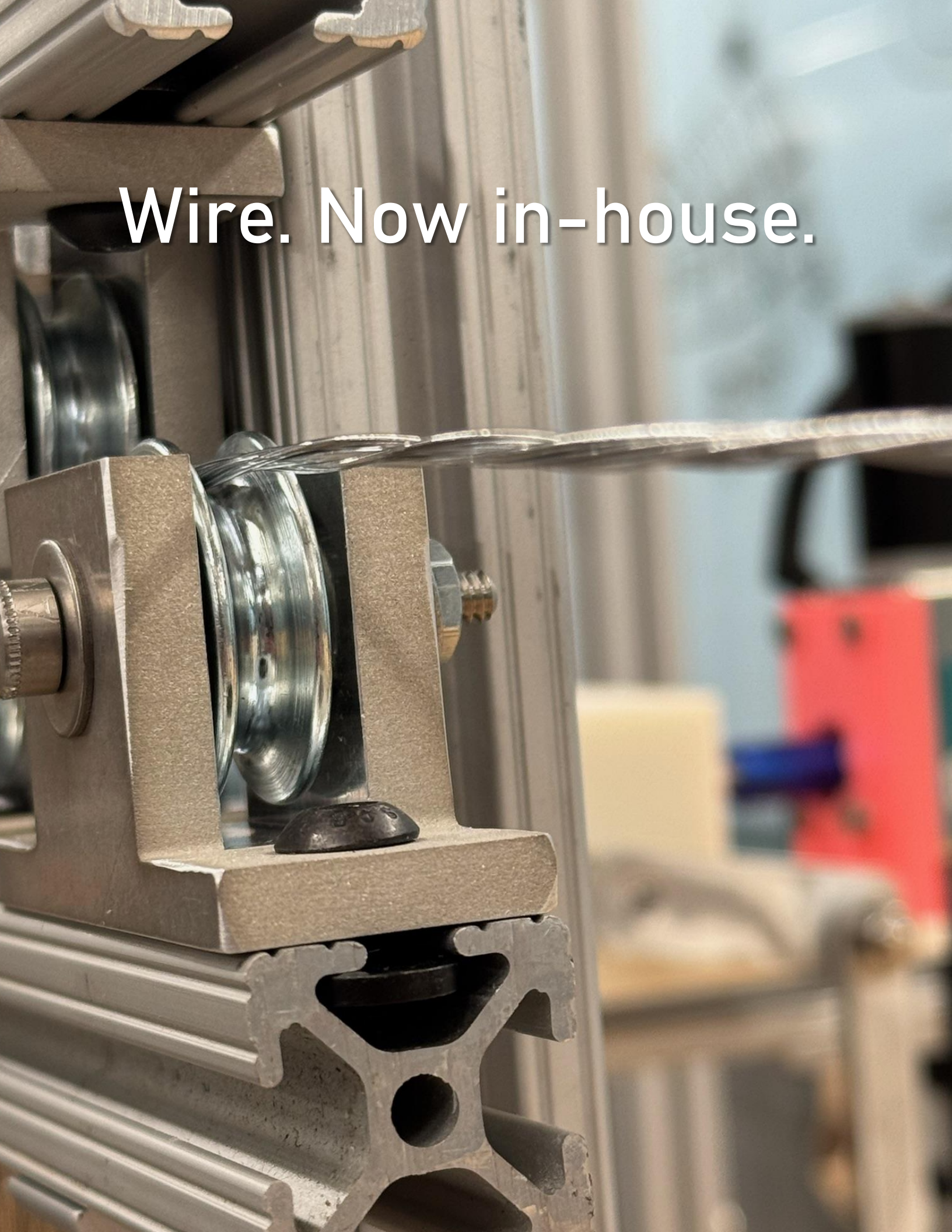


Operation &
Maintenance
Manual

strand.1.0

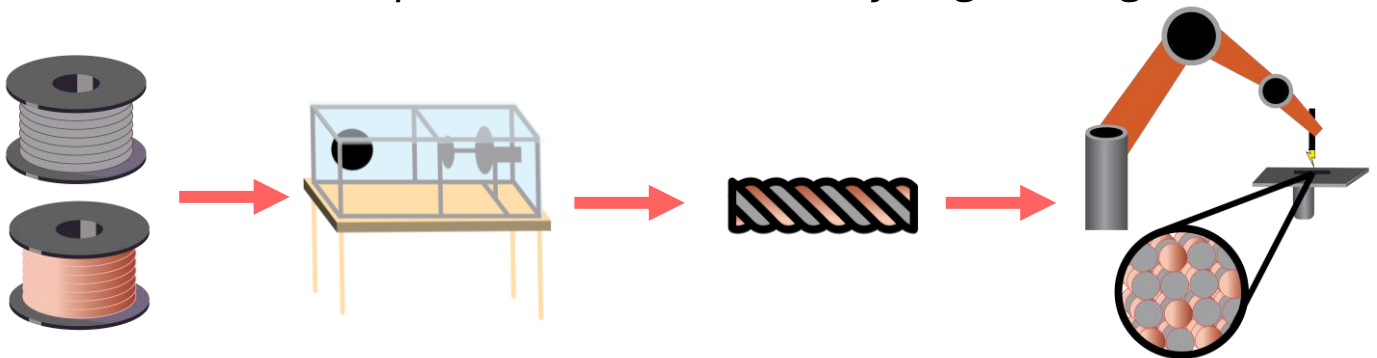
Wire. Now in-house.



IN THIS MANUAL

1. What's Included in Delivery
2. Product Overview
3. User Manual
 - a) Startup
 - b) Operation Guide
 - c) Best Practices
 - d) Repair Guide
 - e) Using your Accessories
 - i. Upgrading to Stronger Motors
4. Bill of Materials
5. Creation Info

strand.1.0, optimized for WAAM alloy engineering



1.

INCLUDED IN
DELIVERY



Included in the delivery of your Wire Strander strand.1.0 are the following:

QUANTITY	ITEM
1	PRE-ASSEMBLED BENCHTOP STRANDING MACHINE
1	1"/300MM DIA STANDARD WELDING/WAAM SPOOL, BLACK
40	FT OF PRE-STRANDED 1 SAMPLE WIRE ON OUTPUT SPOOL MADE OF 1 MILD STEEL, 6 PARTS ALUMINUM
10	2LB SPOOLS OF 30/1000" 4043 ALUMINUM WIRE ON STANDARD 4" WELDING SPOOL (VARYING AMOUNTS ALREADY USED)
10	2LB SPOOLS OF 30/1000" 316L STAINLESS STEEL WIRE ON STANDARD 4" WELDING SPOOL (VARYING AMOUNTS ALREADY USED)
7	2LB SPOOLS OF 30/1000" ER70S-6 MILD STEEL WIRE ON STANDARD 4" WELDING SPOOL (VARYING AMOUNTS ALREADY USED)
1	SHORT ½" KEYED SHAFT FOR INPUT MOTOR
3	SPARE ½" SHAFT COLLARS FOR INPUT MOTOR
1	PLUG FOR BACK OF EMERGENCY STOP BUTTON, USABLE IN MODIFICATIONS
1	PRE-ASSEMBLED WELDING WIRE FEEDER WITH 1.6MM ROLLERS (INCLUDES POWER SUPPLY WIRED TO MOTOR CONTROLLER WIRED TO FEEDER)
1	BLANK ROLLER FOR CUSTOM WIRE DIAMETER, ALUMINUM
8	REPLACEMENT BEARINGS FOR FEEDER
1	BAG OF RING TERMINALS FOR MISCELLANEOUS WIRING REPAIRS
2	DOCYKE 350 KG-CM HIGH TORQUE MOTORS - INTERCHANGEABLE WITH EXISTING MOTORS IN ORDER TO STRAND STRONGER MATERIALS - REQUIRES REWIRING AND MOUNTING

2. PRODUCT OVERVIEW



Introducing the

strand.1.0

the easiest in-house stranding ever



1. INCLUDED

2. OVERVIEW

3. MANUAL

4. CREATION INFO

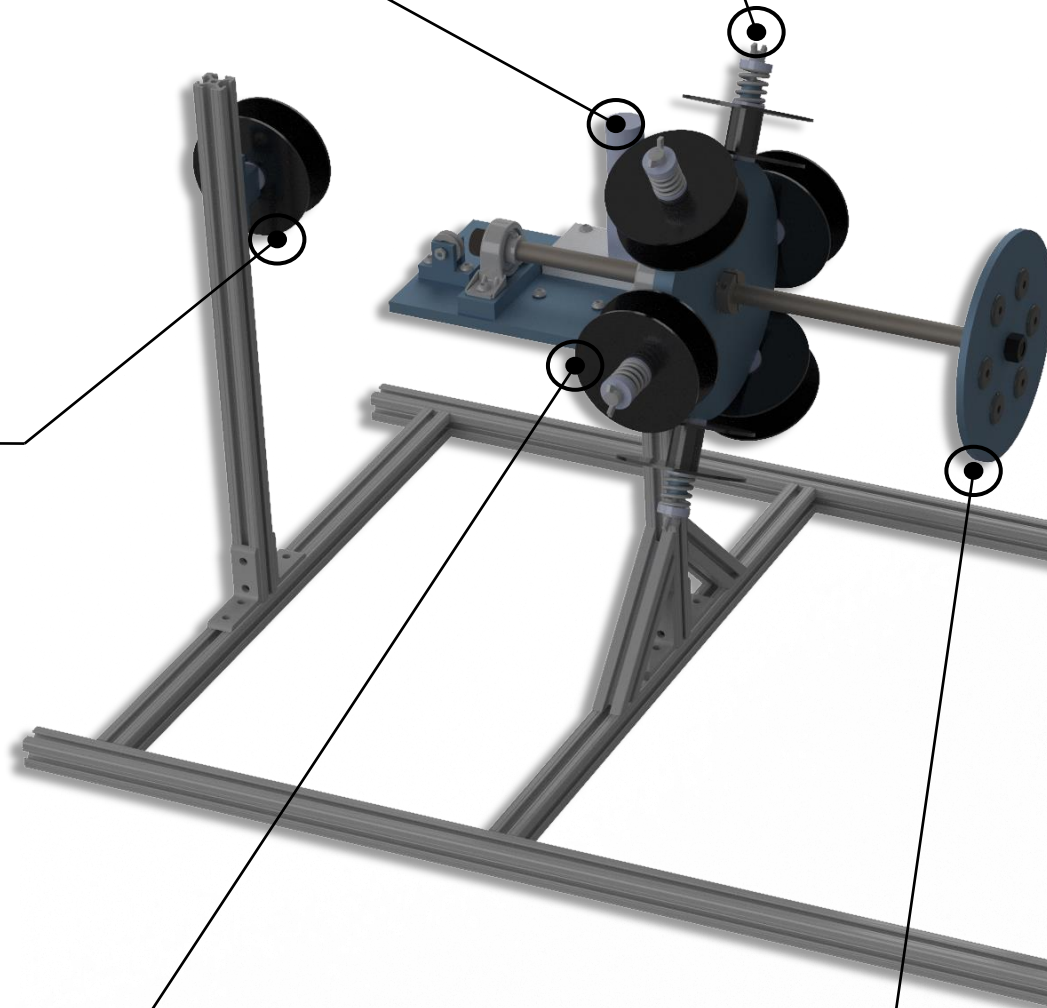
**REMOVABLE
TENSION CAPS**
7 TOTAL

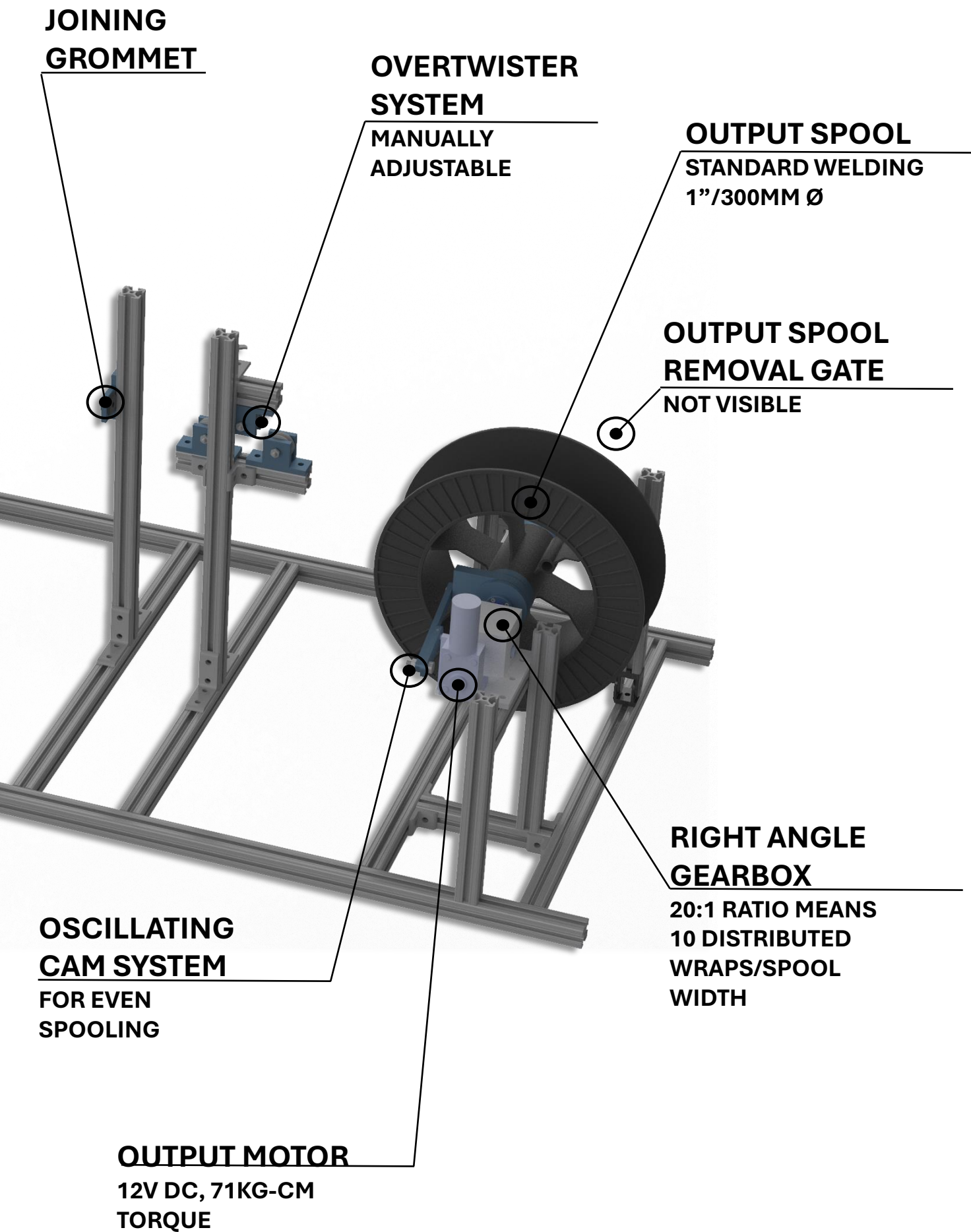
INPUT MOTOR
12V DC, 35KG-CM TORQUE

**INPUT SPOOL
CENTER**

INPUT SPOOLS
SURROUNDING 6

GROMMET WHEEL
WITH REMOVABLE BLACK
RUBBER GROMMETS





OUTPUT MOTOR

12V DC
35KG-CM TORQUE
NOT VISIBLE

POWER BOX

CONTAINS POWER
CONVERTER, MOTOR
CONTROLLERS, RELAY

INPUT MOTOR

12V DC
71KG-CM TORQUE

MAGNETIC
DOOR
LATCHES

DOOR
SENSORS

ALL WIRING
STRAIN
RELIEVED, BUT
CAN BE
DISCONNECTED
AT POWER BOX
END ONLY

ON/OFF

120V AC PLUG

EMERGENCY STOP

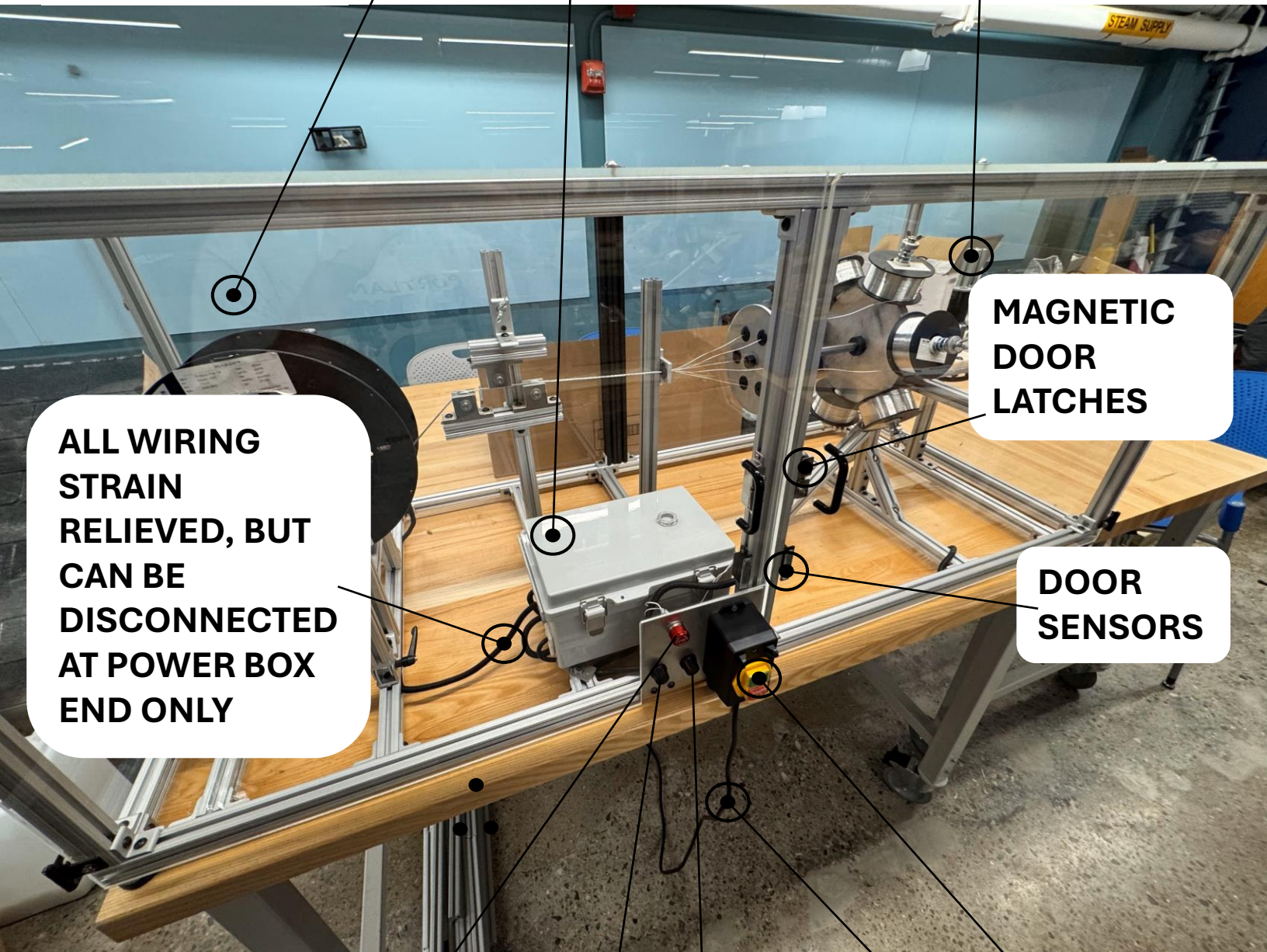
NOT FOR LONG-TERM
POWERING OFF

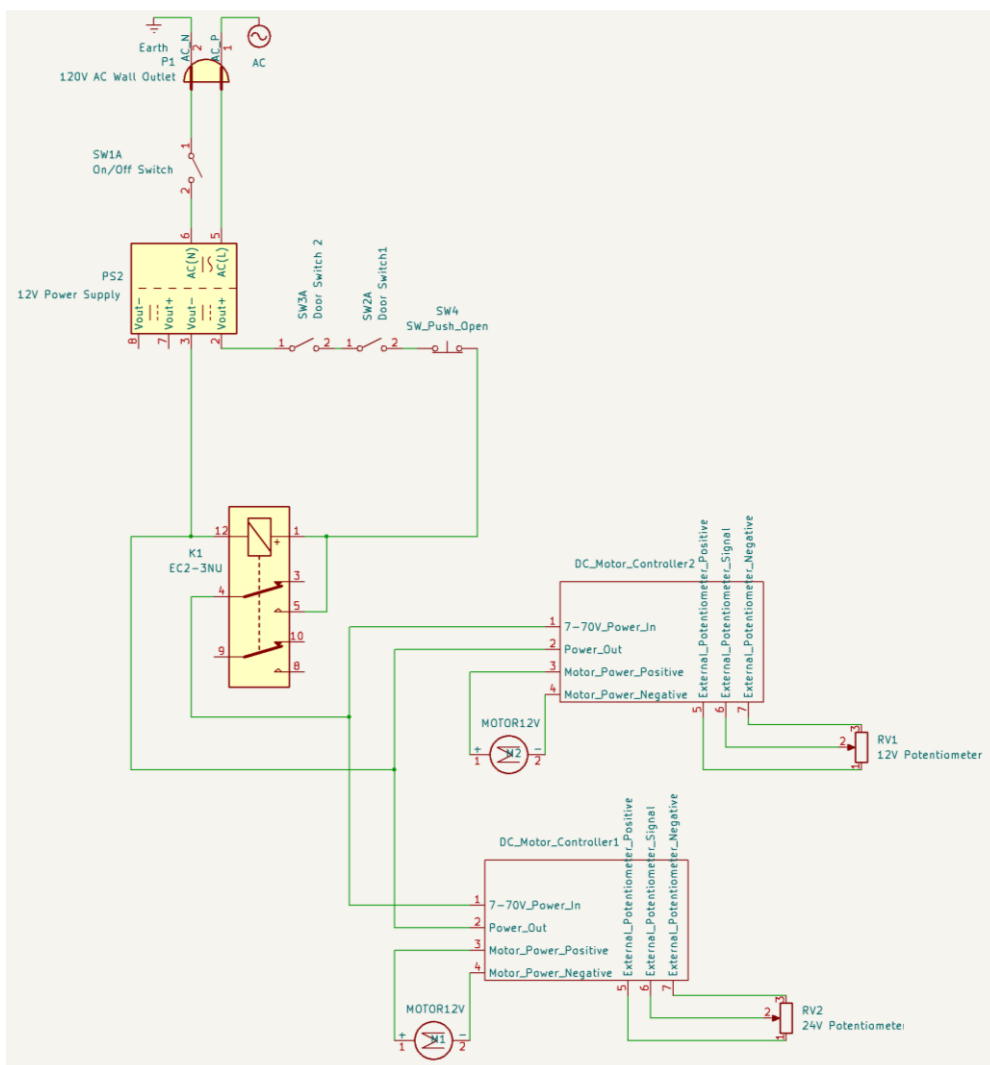
OUTPUT CONTROL

LINEAR SPEED

INPUT CONTROL

PITCH LENGTH





Simplicity and safety

120V AC powers the system, controlled by the on/off button on the control panel. The power is then converted to 12V DC by an off-the-shelf converter, where it passes through a relay that can be interrupted by opening either door or the emergency stop button. Power then passes through the motor controllers and to the motors. That's it. Easy and simple.

Faster stopping when you need it

We optimized user safety by adding the emergency stop to bypass capacitor cooldown that comes from simply turning the machine off. That said, we recommend that you don't use the emergency stop to restrain your strand.1.0 when plugged in as the power supply will still be on and gets hot.

No more pinching

We also implemented the door sensors to safely halt the machine if either door is open. Just be sure not to trip the sensors with other magnets.

3. USER MANUAL



STARTUP

Loading Output Spools



RELEASE LATCHES



FLIP GATE DOWN



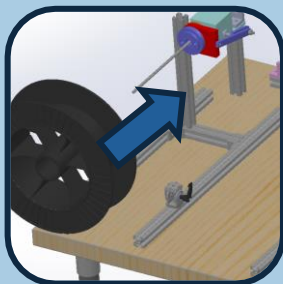
REMOVE M8
SCREWS



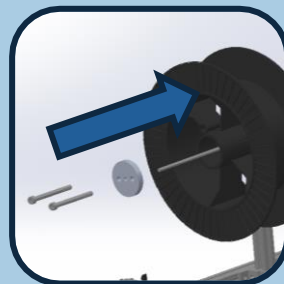
REMOVE END
CAP AND SPOOL



DO NOT REMOVE
FURTHER CAP



PLACE NEW
1" / 300MM SPOOL

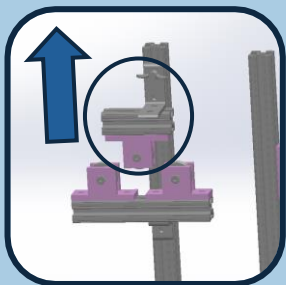


REPLACE ENDCAP
AND SCREW DOWN



FLIP UP GATE AND
TIGHTEN LATCHES

Loading Input Spools & Threading



RAISE OVERTWISTER
PULLEY



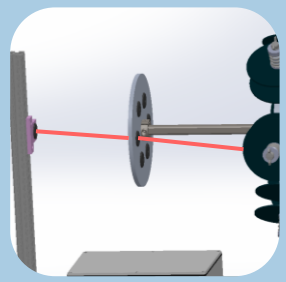
REMOVE
TENSIONER CAP



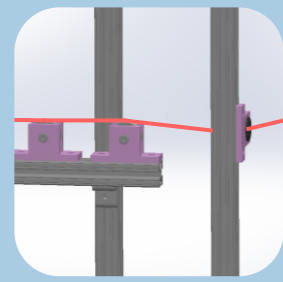
SPOOL UNWINDS
CLOCKWISE



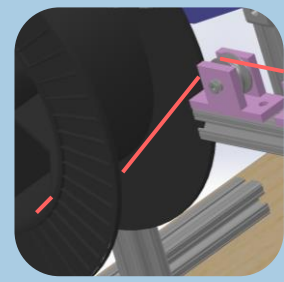
APPLY TENSION
WITH 3 TURNS



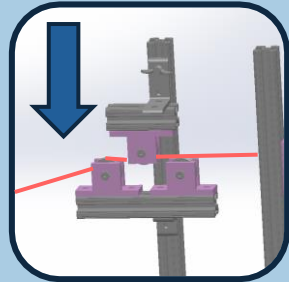
PASS WIRE THROUGH
GROMMETS



PASS WIRE
OVER PULLEYS



HOOK WIRE ON
OUTPUT WHEEL



LOWER OVERTWISTER
& TIGHTEN SCREW

x 7

THAT'S IT! YOU CAN NOW PLUG IN, POWER ON, AND BEGIN STRANDING

1. INCLUDED

2. OVERVIEW

3. MANUAL

4. BOM

5. CREATION INFO

OPERATION GUIDE



7 wire concentric stranding

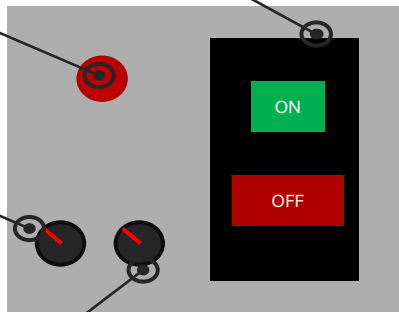
The strand.1.0 creates a “concentric strand” with 1 core wire and 6 surrounding. It could possibly create a 19 wire (double-wrap) strand with modifications. Concentric stranding creates an output diameter of about 3 times the input diameter. The strand.1.0 was designed for .03”/0.8mm input wires, which make a .09”/2.4mm output wire

POWER ON/OFF

EMERGENCY STOP
DOES NOT DISCONNECT POWER
CONVERTER

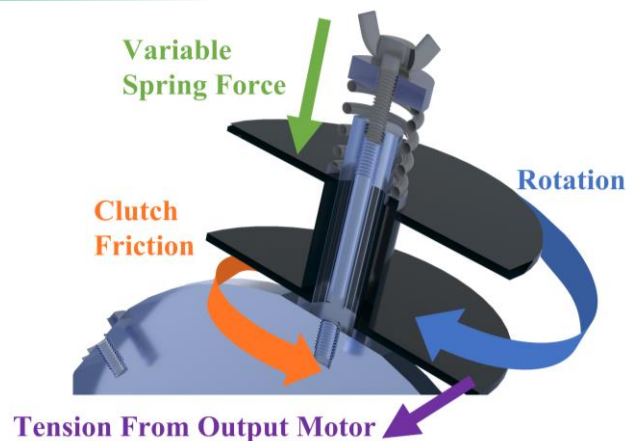
OUTPUT MOTOR SPEED
LINEAR SPEED

INPUT MOTOR SPEED
PITCH LENGTH



Adjustable pitch

By increasing the ratio of the input motor speed and output motor/linear speed, more wraps are created, making for a shorter pitch length. Lengths of about 0.5”/12mm to 2”/50mm can be achieved.



Adjustable tension

By twisting the removable tension caps, tension on the wires can be adjusted. We suggest you begin stranding with 3 full rotations of spring compression

BEST PRACTICES

Input material choice

- Materials with lots of springback are resistant to stranding
- When possible, substitute input materials with less springy ones, for example, using mild steel instead of stainless

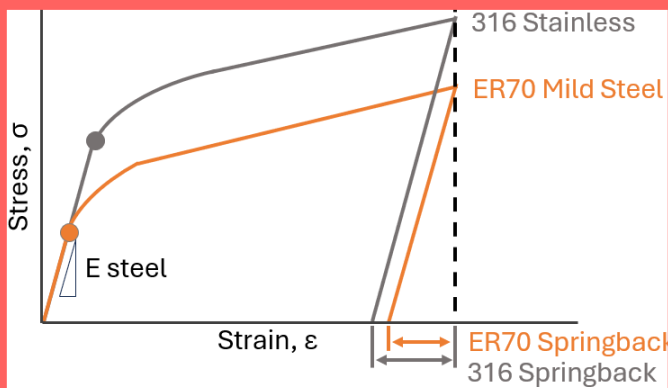
Avoid Harsh Weather, Power Surges, Water

Replace Grommets Often

Allow Motors to Cool

Do Not leave Unattended

Unplug When Not in Use



▲ Among materials with the same E and post-yield behavior, higher yield strength causes more springback

Side Lesson: Springback (Elastic Recovery)

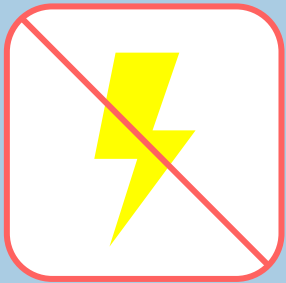
- A material property dependent on modulus of elasticity, E and yield strength
 - In steels, which all have roughly the same E, higher yield strength means more springback
- Caused by different forming methods and compositions
- Combatted by using different materials, more force, or overbending/overtwisting

REPAIR GUIDE

- All framing is ¼-20 screws, driven with 3/16" hex wrench
- Other screws and measurements are a mix of metric and US units due to off-the-shelf availability
- For spare/replacement parts,
 - See Bill of Materials (BoM) for replacement part IDs and access CAD for where they go
 - Custom parts may require machining

Your strand.1.0 wire strander comes with two unattached 12V DC 350kg-cm high-torque motors. Your pre-assembled machine has a 71kg-cm, 200RPM input motor and a 35kg-cm, 5RPM output motor.

Upgrading Input Motor



1. DISCONNECT
POWER PLUG



2. DISCONNECT
COUPLER & GEAR



3. PEEL BACK
COVER AND CUT
BOTH WIRES



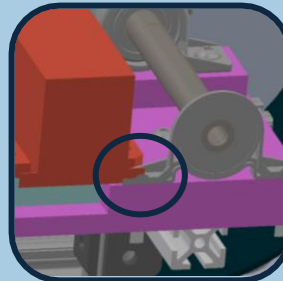
4. CRIMP QUICK
CONNECTS TO
WIRE ENDS



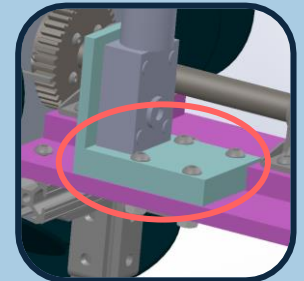
5. REMOVE
EXISTING MOTOR
& MOUNT



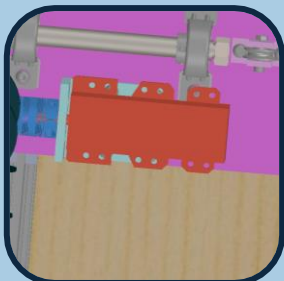
6. CUT BACK
RIGHT FOOT OFF
OF A DOCYKE
MOTOR



7. CUT BACK
PILLOW BEARING
FOOT AS NEEDED



8. PLACE BLOCK
OF ROUGHLY
SAME THICKNESS
(0.602") UNDER
MOTOR



9. DRILL 4.5MM
HOLES THROUGH
BLOCK AND
ALUMINUM PLATE



10. SCREW
DOWN WITH M4
SCREWS



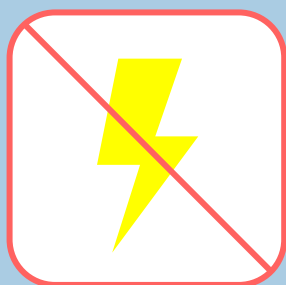
11. REATTACH
COUPLER



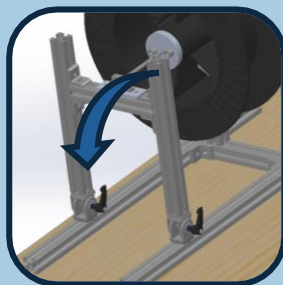
12. ATTACH
WIRES & QUICK
CONNECTS TO
MOTOR

Should you decide to work with stronger wires than the recommended .03" mild steel, you will need to install these motors yourself, following these steps.

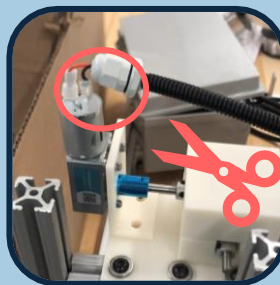
Upgrading Output Motor



1. DISCONNECT
POWER PLUG



2. UNLOAD
OUTPUT SPOOL



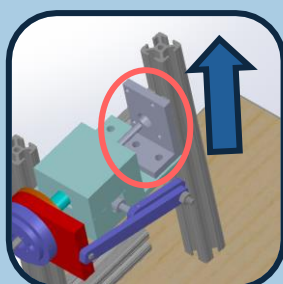
3. PEEL BACK
COVER AND CUT
BOTH WIRES



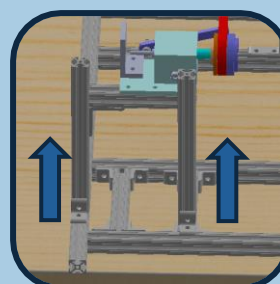
4. DISCONNECT
COUPLER AND
REMOVE MOTOR



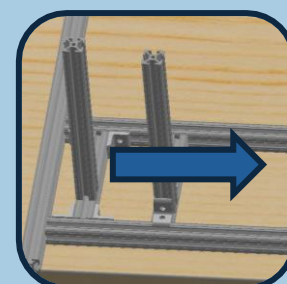
5. CRIMP QUICK
CONNECTS TO
WIRE ENDS



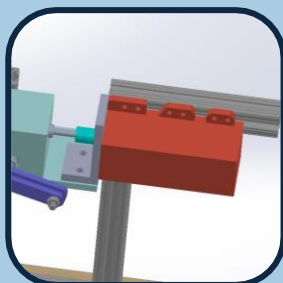
6. REMOVE
MOTOR MOUNT



7. UNSCREW
BOTH UPRIGHTS
(DOCYKE
MOTOR NEEDS
MORE SPACE)



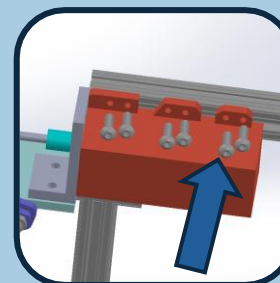
8. REATTACH
UPRIGHTS TO
CROSSBAR RAILS,
MOVES
EVERYTHING
TOWARDS DOORS



9. ATTACH 80/20
RAIL FOR
MOUNTING



10. RECONNECT
COUPLER



11. ATTACH MOTOR
WITH M4 SCREWS
(SEE MOUNTING OF
MAGNETIC DOOR
LATCH)



12. ATTACH
WIRES & QUICK
CONNECTS TO
MOTOR

4.

BILL OF MATERIALS

For an electronic version of this list or for a CAD model of all parts, contact

s.Julien@northeastern.edu

or visit

<https://docs.google.com/spreadsheets/d/1jxrDtaIZwZ8xc4BR9ZYKxwxKtzseA7Ym3rSpSmJ2HA/edit?usp=sharing>



1. INCLUDED

2. OVERVIEW

3. MANUAL

▶ **4. BOM**

5. CREATION INFO



Subsystem	CAD subsystem	CAD part name	Description	Part quantity needed	Part number	Part name + link	Cost Each	Cost Total	BoM PAGE 1 OF 12
Electrical	N/A	N/A	Power supply	1	B0D6V6KGHK	12V Power Supply	\$39.99	\$39.99	
Electrical	N/A	N/A	Gray Electronics Housing box	1	B08282SQPT	Protective Electrical Box	\$35.49	\$35.49	
Electrical	N/A	N/A	Emergency Stop	1	B0C9QD72SL	Emergency Stop 120V Power Cord	\$28.99	\$28.99	
Electrical	N/A	N/A	DPDT	1	B07FCJ39QR	DPDT Relay - 12V coil option	\$21.99	\$21.99	
Electrical	N/A	N/A	12V Motor Controller	2	B071NQ5G71	RioRand 7-70V PWM DC Motor Speed Controller Switch 30A	\$15.99	\$0.00	
Electrical	N/A	N/A	Cable covering	1	B0CX8YQG7H	25 1/4" Cable Covering	\$8.99	\$8.99	
Electrical	N/A	N/A	Wire strain relief	7	B094ZDSD86	1/4" Strain relief glands	\$8.66	\$60.62	
Electrical	N/A	N/A	Door switch	3	B0B3D7BM4K	Door Reed Switch	\$7.99	\$23.97	
Electrical	No CAD available	No CAD available	reed switches	1	B086GYJLML	2pack black reed switches NO,NC	\$6.37	\$6.37	
Electrical	N/A	N/A	Emergency Stop Button	1	B09MTMNP61	Emergency Stop Button	\$11.99	\$11.99	
Feeder for Testing	For future builds	For future builds	Ball bearings for welding feeder wheels	3	B07FW8P57M	10mmx19mm bearings for Welding feeder rig	\$8.29	\$24.87	
Feeder for Testing	N/A	N/A	Feeder	1	SSJ-17	https://www.amazon.sa/-/en/Welding-Feeding-Assembly-1-2-3-2mm-047-126/dp/B07D35HF64	\$218.35	\$218.35	

Feeder for Testing	N/A	N/A	Power Supply for Feeder	1	B0CP9HP93V	AC to DC Converter 60V DC Power Supply	\$63.99	\$63.99
Feeder for Testing	N/A	N/A	12V Motor Controller for Feeder	1	RR-MJLK-FBDG	RioRand 7-70V PWM DC Motor Speed Controller Switch 30A	\$13.99	\$13.99
Frame	20241012-STRANDER-MACHINABLE	47065T413_T-Slotted Framing	5 ft 8020	2	N/A	N/A	\$0.00	\$0.00
Frame	Miscellaneous build	No CAD available	1/4-20, 5/8" length	3	97763A264	Stainless Button Head Hex Screw, 1/4"-20, 5/8" Long, pack of 25	\$6.57	\$19.71
Frame	No CAD available	No CAD available	Rubber feet	1	9540K36	Unthreaded Bumper, 1-1/2" OD, pack of 10	\$10.84	\$10.84
Frame	Shielding	No CAD available	To assemble framing	2	B0D3DK2K5F	8020 L bracket 10 pk	\$20.99	\$41.98
Frame	Shielding	No CAD available	To assemble framing	2	B0CQY7B262	8020 Nuts + screws	\$9.99	\$19.98
Frame	Shielding	12097_m_a & 12097_m_b & 12097_m_pin		4	12097	Hinge	\$8.00	\$32.00
Frame	Shielding	No CAD available	to mount magnetic latches	4	13082	M3 8020 nut	\$1.74	\$6.96
Frame	Shielding	No CAD available	Hinge mounting	8	3205	10-32 8020 nut	\$1.16	\$9.28
Frame	Shielding	8020 2 ft	original 8020 order	10	N/A	N/A	\$0.00	\$0.00
Frame	Shielding	8020 5 ft	original 8020 order	4	N/A	N/A	\$0.00	\$0.00

Subsystem	CAD subsystem	CAD part name	Description	Part quantity needed	Part number	Part name + link	Cost Each	Cost Total	BoM PAGE 3 OF 12
Frame	Shielding	8020 25 in	original 8020 order	4	N/A	N/A	\$0.00	\$0.00	
Frame	Shielding	36 x 24 Plexi	Top/Back Panels	4	N/A	N/A	\$0.00	\$0.00	
Frame	Shielding	25 x 24 Plexi	Side Panels	2	N/A	N/A	\$0.00	\$0.00	
Frame	Shielding	Left swinging door	Left door	1	N/A	N/A	\$0.00	\$0.00	
Frame	Shielding	right swinging door	Right door	1	N/A	N/A	\$0.00	\$0.00	
Frame	Shielding	92141A011_18-8 stainless steel washer	washers	8	N/A	N/A	\$0.00	\$0.00	
Frame	Shielding	92210A302_18-8 Stainless Steel Hex Drive Flat Head Screw	screws	2	N/A	N/A	\$0.00	\$0.00	
Frame	Shielding	91841A195_18-8 Stainless Steel Hex Nut	nuts	3	N/A	N/A	\$0.00	\$0.00	
Frame	Shielding	92210A197_18-8 Stainless Steel Hex Drive Flat Head Screw	screws	2	N/A	N/A	\$0.00	\$0.00	
Input	20241011-spool wheel - v2	2464K32_Clamping Precision Flexible Shaft Coupling	Shaft coupling for motor shaft and keyed shaft	1	2464K32	Shaft coupling	\$101.62	\$101.62	
Input	20241011-spool wheel - v2	gear-screw-machined	Input gears - screw	1	6325K19	gears, machinable bore	\$62.63	\$62.63	

Input	20241011-spool wheel - v2	gear-keyed-machined	Input gears - keyed	1	6325K19		gears, machinable bore	\$62.63	\$62.63
Input	20241011-spool wheel - v2	20241022-hollow d shaft	Steel input shaft	1	89955K829		Shaft	\$42.01	\$42.01
Input	20241011-spool wheel - v2	20241022-hollow d shaft	Aluminum input shaft	1	9056K91-9056K912		aluminum shaft, 2 ft long, Aluminum Tube	\$16.02	\$16.02
Input	20241011-spool wheel - v2	9307K871_Buna-N Rubber Grommets	grommets	1	9307K871		Grommet, pack of 10	\$13.94	\$13.94
Input	20241011-spool wheel - v2	No CAD available	1/4-20 1 1/2" length	4	97763A348		1/4-20 screws, 1 1/2 length, button head, pack of 25	\$11.43	\$45.72
Input	20241011-spool wheel - v2	97763A428_18-8 Stainless Steel Button Head Hex Drive Screws	Screws for mounting gear to wheel	4	97763A428		m4 screws, 35mm length, pack of 10	\$11.00	\$44.00
Input	20241011-spool wheel - v2	5913K62_Low-Profile Mounted Sealed Steel Ball Bearing	Input bearings for shaft	2	5913K62		bearings	\$10.95	\$21.90
Input	20241011-spool wheel - v2	No CAD available	Input shaft coupler	1	B098NLNHR5		1pack 8mm-.5" rigid coupler- clamp screws	\$10.49	\$10.49
Input	20241011-spool wheel - v2	6436K15_Clamping Two-Piece Shaft Collar	input shaft collar	2	6436K15		Collar for shaft	\$10.04	\$20.08
Input	20241011-spool wheel - v2	No CAD available	Input shaft coupler	1	B0B2PWQL6Z		10pack 6mm-8mm rigid coupler- set screws	\$9.99	\$9.99
Input	20241011-spool wheel - v2	No CAD available	10-32, 9/16" length	1	91255A027		10-32 screws, 9/16" length, pack of 25	\$9.76	\$9.76
Input	20241011-spool wheel - v2	95479A122_Medium-strength steel hex nut	Input shaft nut Screws for	2	95479A122		Nut for shaft, pack of 10	\$9.23	\$18.46
Input	20241011-spool wheel - v2	92949A836_18-8 Stainless Steel Button Head Hex Drive Screw	mounting bearings and motor mount	8	92949A836		1/4-20 screws, 1 3/8 length, pack of 25	\$9.20	\$73.60

Subsystem	CAD subsystem	CAD part name	Description	Part quantity needed	Part number	Part name + link	Cost Each	Cost Total	BoM PAGE 5 OF 12
Input	20241011-spool wheel - v2	1497K7_1045 Carbon Steel Keyed Rotary Shaft	Input gear keyed shaft	1	1497K7	Keyed Rotary shaft	\$8.97	\$8.97	
Input	20241011-spool wheel - v2	97022A878_316 Stainless Steel Ring Shim	Smaller pulley shim	2	97022A878	.188 IN ID, .25 IN OD, .012 IN THICKNESS, STEEL SHIM, FOR SMALLER PULLEY	\$7.43	\$14.86	
Input	20241011-spool wheel - v2	97381A207_Aluminum Shoulder Screw	Smaller pulley shoulder screw	1	97381A207	3/4 IN SHOULDER LENGTH, 8-32 THREAD, 3/16 IN DIAMETER, FOR SMALLER PULLEY	\$7.37	\$7.37	
Input	20241011-spool wheel - v2	97763A264_18-8 Stainless Steel Button Head Hex Drive Screws	Screws for mounting center spool mount	2	97763A264	1/4-20 screws, 5/8 length, pack of 25	\$6.57	\$13.14	
Input	20241011-spool wheel - v2	98164A911_316 Stainless Steel Button Head Hex Drive Screw	Screws for smaller pulley frame mount	2	98164A911	1/4-20 screws, 1 1/8 length, pack of 10	\$5.77	\$11.54	
Input	20241011-spool wheel - v2	3434T74_Pulley for Wire Rope	Smaller pulley	1	3434T74	1/8 IN DIAMETER, 1 IN OD, FOR SMALLER PULLEY	\$5.17	\$5.17	
Input	20241011-spool wheel - v2	92625A416_Stainless Steel Wing-Head Thumb Screw	winghead thumb screw for input tensioning	7	92625A416	Stainless Steel Wing-Head Thumb Screw	\$4.99	\$34.93	
Input	20241011-spool wheel - v2	joining-grommet-machinable	Joining grommet	1	N/A	N/A	\$1.39	\$1.39	
Input	20241011-spool wheel - v2	INPUT ASSEMBLY - MOUNT	Smaller pulley mount with clearance holes	1	N/A	N/A	\$0.00	\$0.00	
Input	20241011-spool wheel - v2	20240926-center-spool-mount-v3	Center spool mount with clearance holes	1	N/A	N/A	\$0.00	\$0.00	
Input	20241011-spool wheel - v2	INPUT ASSEMBLY - MOUNT	Bearing height adjustment with clearance holes	2	N/A	N/A	\$0.00	\$0.00	
Input	20241011-spool wheel - v2	INPUT ASSEMBLY - MOUNT	Motor mount with clearance holes	1	N/A	N/A	\$0.00	\$0.00	

Input	20241011-spool wheel - v2	INPUT ASSEMBLY - MOUNT	Input subassembly mount	1	N/A	N/A	\$0.00	\$0.00
Input	20241011-spool wheel - v2	20240926-Grommet Wheel	Grommet wheel	1	N/A	N/A	\$0.00	\$0.00
Input	20241011-spool wheel - v2	7th wheel axle	wheel axles	7	N/A	N/A	\$0.00	\$0.00
Input	20241011-spool wheel - v2	tension friction disk	tension friction disk	7	N/A	N/A	\$0.00	\$0.00
Input	20241011-spool wheel - v2	tensioner cap	tensioner cap	7	N/A	N/A	\$0.00	\$0.00
Input	20241011-spool wheel - v2	1 inch 80-20 4 Slot (1)	4 in 8020	1	N/A	N/A	\$0.00	\$0.00
Input	20241011-spool wheel - v2	47065T209_T-Slotted Framing	22 in 8020	4	N/A	N/A	\$0.00	\$0.00
Input	20241011-spool wheel - v2	47065T554_T-Slotted Framing	18 in 8020	2	N/A	N/A	\$0.00	\$0.00
Input	20241011-spool wheel - v2	47065T553_T-Slotted Framing	6 in 8020	1	N/A	N/A	\$0.00	\$0.00
Input	20241011-spool wheel - v2	47065T411_T-Slotted Framing	10 in 8020	1	N/A	N/A	\$0.00	\$0.00
Input	20241011-spool wheel - v2	47065T239_Silver Corner Bracket	long right angle brackets	9	N/A	N/A	\$0.00	\$0.00
Input	20241011-spool wheel - v2	20240926-pulley-frame	smaller pulley frame Screws for	1	N/A	N/A	\$0.00	\$0.00
Input	20241011-spool wheel - v2	92949A540_18-8 Stainless Steel Button Head Hex Drive Screw	mounting input subassembly base plate	3	N/A	N/A	\$0.00	\$0.00

Subsystem	CAD subsystem	CAD part name	Description	Part quantity needed	Part number	Part name + link	Cost Each	Cost Total	BoM PAGE 7 OF 12
Input	20241011-spool wheel - v2	95505A601_Medium-strength steel hex nut	Nuts for input subassembly plate screws	8	N/A	N/A	\$0.00	\$0.00	
Input	20241011-spool wheel - v2	97763A232_18-8 Stainless Steel Button Head Hex Drive Screws	Set Screw for d shaft and gear	1	N/A	N/A	\$0.00	\$0.00	
Input	20241011-spool wheel - v2	93181A009_Aluminum hex nut	Nut for smaller pulley shoulder screw	1	N/A	N/A	\$0.00	\$0.00	
Input	For future builds	For future builds	Rigid coupler for input motor	1	B098NLNHR5	8mm to 12.7mm coupler for input motor	\$10.49	\$10.49	
Input	For future builds	For future builds	1/2" shaft collar	1	1/2-COSC-4P	Shaft collar for 0.5" shaft diameter	\$8.99	\$8.99	
Input	N/A	N/A	200RPM Input Motor	1	B08FFN3DS2	12V 200RPM 71kgcm motor	\$39.99	\$39.99	
Input	pulley mechanism	92625A416_Stainless Steel Wing-Head Thumb Screw	Tensioning screws	7	92625A418	Winghead thumb screw	\$2.47	\$17.29	
Input	pulley mechanism	90480A011_Low-Strength Steel Hex Nut	nut for larger pulley shoulder screw	3	90480A011	Nuts for shoulder screw, pack of 100	\$2.33	\$6.99	
Input	20241011-spool wheel - v2	98870A110_Machine Key	Key for input gear keyed shaft	1	98870A110	Machine Key	\$3.23	\$3.23	
Input	20241011-spool wheel - v2	inputspool-wheel-v3	Input spool wheel	1	N/A	N/A	\$0.00	\$0.00	
Output	20241012-STRANDER-MACHINABLE	NEW-INPUT-MOTOR	Output motor	1	B07YBXB4N7	12V, 5rpm, 35kgcm motor	\$29.99	\$29.99	
Output	20241012-STRANDER-MACHINABLE	921292A020_18-8 Stainless Steel Socket Head Screw	Spline nut screws	4	91292A020	M3 x 0.5 mm Thread, 25 mm Long	\$8.72	\$34.88	

Output	3-output-subsystem	gearbox-assembly	Output gearbox	1	TYXC0016_1:20_6	20:1 Right Angle Worm Gearbox 90° Angle	\$41.99	\$41.99
Output	3-output-subsystem	47065T221_T-Slotted Framing	Locking pivot for 8020	2	47065T221	Locking Pivot for 1" High Single Rail	\$23.37	\$46.74
Output	3-output-subsystem	1865K36 Aluminum Easy-Access Base-Mounted Shaft Support	Shaft support for output shaft	1	1865K36	ALUMINUM EASY-ACCESS BASE-MOUNTED SHAFT SUPPORT	\$17.46	\$17.46
Output	3-output-subsystem	90269A258_316 Stainless Steel Shoulder Screw	clamps output flywheel and arm	1	90269A258	316 Stainless Steel Shoulder Screw	\$14.78	\$14.78
Output	3-output-subsystem	gearbox-assembly	Shaft couplings for output subassembly	2	AS015C	6Pack Treehobby Aluminum Alloy Shaft Coupling Rigid	\$8.97	\$17.94
Output	3-output-subsystem	91292A520_18-8 Stainless Steel Socket Head Screw	Output spool clamping screws	2	91292A520	M8 x 1.25 mm Thread, 110 mm Long	\$8.37	\$16.74
Output	3-output-subsystem	94128A118_Headless Shoulder Screws	engages with output slider	1	94128A118	Headless Shoulder Screws	\$6.86	\$6.86
Output	3-output-subsystem	N/A	sliding spool release thumb screws	1	99607A234	1/4-20 Thumb Screw	\$5.00	\$5.00
Output	3-output-subsystem	92605A100_Stainless Steel Flat-Tip Set Screw	set screw for output flywheel	1	92605A100	Stainless Steel Flat-Tip Set Screw	\$3.50	\$3.50
Output	3-output-subsystem	3-arm	3-arm	1	N/A	N/A	\$0.00	\$0.00
Output	3-output-subsystem	3-flywheel	3-flywheel	1	N/A	N/A	\$0.00	\$0.00
Output	3-output-subsystem	slider 2	slider 2	1	N/A	N/A	\$0.00	\$0.00
Output	3-output-subsystem	Right spool cap spacer	Right spool cap spacer	1	N/A	N/A	\$0.00	\$0.00

Subsystem	CAD subsystem	CAD part name	Description	Part quantity needed	Part number	Part name + link	Cost Each	Cost Total	BOM PAGE 9 OF 12
Output	3-output-subsystem	Right spool cap	Right spool cap	1	N/A	N/A	\$0.00	\$0.00	
Output	3-output-subsystem	Left spool cap	Left spool cap	1	N/A	N/A	\$0.00	\$0.00	
Output	3-output-subsystem	20240926-shaft_mount_base	Shaft Mount Base	1	N/A	N/A	\$0.00	\$0.00	
Output	3-output-subsystem	3-ROLO DE ARAME DE SOLDA	5lb output spool	1	N/A	N/A	\$0.00	\$0.00	
Output	3-output-subsystem	20240926-shaft_mount_base	shaft support base	1	N/A	N/A	\$0.00	\$0.00	
Output	3-output-subsystem	MOTOR-AND-GEARBOX-MOUNT-GEARBOX-MOUNT-BOTTOM	gearbox mount bottom	1	N/A	N/A	\$0.00	\$0.00	
Output	3-output-subsystem	MOTOR-AND-GEARBOX-MOUNT-GEARBOX-MOUNT-TOP	gearbox mount top	1	N/A	N/A	\$0.00	\$0.00	
Output	3-output-subsystem	MOTOR-AND-GEARBOX-MOUNT-MOTOR MOUNT	motor mount	1	N/A	N/A	\$0.00	\$0.00	
Output	3-output-subsystem	47065T411_T-Slotted Framing	10 in 8020	4	N/A	N/A	\$0.00	\$0.00	
Output	3-output-subsystem	47065T553_T-Slotted Framing	6 in 8020	3	N/A	N/A	\$0.00	\$0.00	
Output	3-output-subsystem	47065T209_T-Slotted Framing	22 in 8020	2	N/A	N/A	\$0.00	\$0.00	
Output	20241012-STRANDER-MACHINABLE	921292A020_18-8 Stainless Steel Socket Head Screw	Splined nut screws	4	91292A020	M3 x 0.5 mm Thread, 25 mm Long	\$8.72	\$34.88	

Output	3-output-subsystem	3-47065T831_Black Corner Bracket	short right angle brackets	8	N/A	N/A			\$0.00	\$0.00
Output	3-output-subsystem	47065T239_Silver Corner Bracket	long right angle brackets	6	N/A	N/A			\$0.00	\$0.00
Output	3-output-subsystem	91828A251_18-8 Stainless Steel Hex Nut	M6 Nut Right angle gearbox to accommodate larger motor	1	N/A	N/A			\$0.00	\$0.00
Output	For future builds	For future builds		1	B09WB1X7BY	Right angle gearbox (20:1 option)			\$41.99	\$41.99
Output	3-output-subsystem	61145K137_Splined Shaft	Spline shaft for output spool subassembly	1	61145K137	Steel Ball Spline			\$154.37	\$154.37
Output	3-output-subsystem	61145K41_Mounted Linear Steel Ball Bearing for 6mm Diameter Splined Shaft	Bearing for output spool subassembly	1	61145K41	Mounted Linear Steel Ball Bearing for 6 mm Diameter Steel Ball Spline			\$97.01	\$97.01
Output & Input	For future builds	For future builds	Higher torque input and output motors	2	B09XDXSFRR	DOCYKE motor (350kg option)			\$71.99	\$143.98
Output & Input	N/A	N/A	12V Motor Controller	2	RIORAND, RR-MJLK-FBDG	RioRand 7-70V PWM DC Motor Speed Controller Switch 30A			\$13.99	\$27.98
Overtwister	20241012-STRANDER-MACHINABLE	47065T239_Silver Corner Bracket	long right angle brackets	2	N/A	N/A			\$0.00	\$0.00
Overtwister	pulley mechanism	No CAD available	1/4-20 5/8" length	6	97763A264	1/4-20 screws, 5/8 length, button head, pack of 25			\$0.00	\$0.00
Overtwister	pulley mechanism	3434T23_Pulley for Wire Rope	bigger pulley	2	3434T23	1-1/4 Pulley for Wire Rope			\$6.25	\$12.50
Overtwister	pulley mechanism	3434T23_Pulley for Wire Rope	bigger pulley	1	3434T23	3/16 INCH DIAMETER, 1.25 IN OUTER DIAMETER, FOR LARGER PULLEY			\$6.25	\$6.25
Overtwister	pulley mechanism	91404A537_Zinc-plated Iron Wing-Head Thumb Screw	screw for pulley mechanism	1	91404A537	1/4-20 Wing-Head Thumb Screw			\$4.11	\$4.11

Subsystem	CAD subsystem	CAD part name	Description	Part quantity needed	Part number	Part name + link	Cost Each	Cost Total	BOM PAGE 11 OF 12
Overtwister	pulley mechanism	91259A542_Alloy Steel Shoulder Screw	Larger pulley shoulder screw	3	97832A154	1.125 IN SHOULDER LENGTH, 1/4 IN SHOULDER DIAMETER, 10-24 THREAD, FOR LARGER PULLEY, LONGER OPTION	\$3.36	\$10.08	
Overtwister	pulley mechanism	8020-overtwister	18 in 8020	1	N/A	N/A	\$0.00	\$0.00	
Overtwister	pulley mechanism	5 in 8020]	5 in 8020	1	N/A	N/A	\$0.00	\$0.00	
Overtwister	pulley mechanism	3 in 8020	3 in 8020	1	N/A	N/A	\$0.00	\$0.00	
Overtwister	pulley mechanism	3-47065T831_Black Corner Bracket	short right angle brackets	3	N/A	N/A	\$0.00	\$0.00	
Overtwister	pulley mechanism	47065T239_Silver Corner Bracket	long right angle brackets	1	N/A	N/A	\$0.00	\$0.00	
Overtwister	pulley mechanism	20240926-bigger-pulley-frame	bigger pulley frame	3	N/A	N/A	\$0.00	\$0.00	
Overtwister	pulley mechanism	No CAD available	pulley mechanism mounting screws	6	N/A	N/A	\$0.00	\$0.00	
Overtwister	N/A	N/A	Larger pulley mount with clearance holes	1	N/A	N/A	\$0.00	\$0.00	
Safety	2 Horizontal Doors Final	1101A12_Surface Mount Magnetic Latch	Shield door latch	2	1101A12	Door latch	\$4.69	\$9.38	
Output	3-output-subsystem	47065T209_T-Slotted Framing	22 in 8020	2	N/A	N/A	\$0.00	\$0.00	
Output	20241012-STRANDER-MACHINABLE	921292A020_18-8 Stainless Steel Socket Head Screw	Splined nut screws	4	91292A020	M3 x 0.5 mm Thread, 25 mm Long	\$8.72	\$34.88	

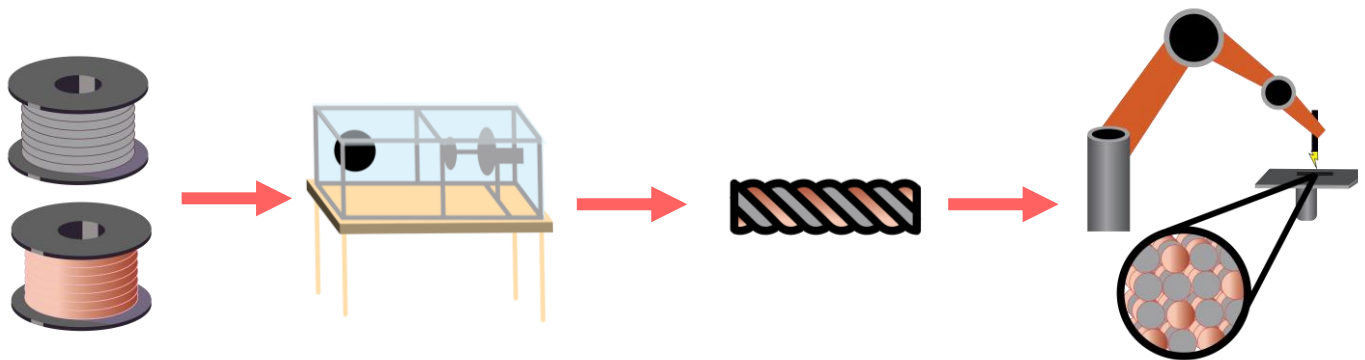
Safety	2 Horizontal Doors Final	11945A11_Unthreaded-Hole Round Pull Handle	Shield door handle	2	11945A11	Pull handle	\$0.86	\$1.72
Safety	N/A	N/A	Sharp objects sticker	5	B00SY11WX4	Sharp Objects Sticker	\$10.08	\$50.40
Safety	N/A	N/A	Safety glasses sticker	1	B01E1DFRV8	Safety Glasses Sticker	\$3.99	\$3.99
Safety	Shielding	92196A271_18-8 Stainless steel socket head screw	Screws for shielding	5	92196A271	10-32 Socket Head Cap Screw	\$15.05	\$75.25
Safety	N/A	N/A	Pinch point sticker	4	769152879215	Pinch Point Sticker 4pk	\$6.99	\$27.96
Tensioning	20241011-spool wheel - v2	No CAD available	To hold together spring + friction disks	1	B0D69N9623	Double-sided tape	\$9.99	\$9.99
Tensioning	20241011-spool wheel - v2	9657K325_Compression Spring	Spring for tensioning V2, pack of 6	12	9657K319	Compression Spring	\$7.39	\$88.68
Tensioning	20241011-spool wheel - v2	9657K325_Compression Spring	Spring for tensioning V1	7	9657K325	Compression Spring	\$5.48	\$38.36
Tensioning	pulley mechanism	No CAD available	M6 screws x 30 screws, pack of 6	6	91292A139	M6 x 1 mm Thread, 30 mm Long	\$12.46	\$74.76
Tensioning	pulley mechanism	No CAD available	M6 x 25 screws, pack of 6	6	91292A138	M6 x 1 mm Thread, 25 mm Long	\$11.99	\$71.94
Tensioning	pulley mechanism	91259A542_Alloy Steel Shoulder Screw	Tensioning pulley shoulder screws	3	97832A154	Larger pulley shoulder screws	\$3.36	\$10.08
Wire	20241011-spool wheel - v2	spool	2 lb steel spools	2	B	5 Rolls – Stainless Mig welding wire 316L .030” X 2 lb	\$77.80	\$155.60
Wire	20241011-spool wheel - v2	spool	1 LB aluminum spools	2	A	5 x 1 LB Aluminum 4043 MIG Welding Wire ER4043 .030” (0.8mm) 1 LB 5 PK	\$42.54	\$85.08

5. PRODUCT CREATION



INTENDED APPLICATION

This wire strander was designed to create twisted cable feedstock for use with Wire Arc Additive Manufacturing (WAAM). By twisting together various metals, alloys can be formed during deposition that cannot be commercially purchased as wire for feedstock. In particular, the near-equal concentrations are good for making High Entropy Alloys (HEA's). This machine was designed to interface with a Fronius/ABB WAAM cell's feeder, tubing, and plasma torch.



PROPERTY RIGHTS

This machine was created as an undergraduate senior engineering capstone project in the Northeastern University Mechanical and Industrial Engineering Department. The team of students that has created and owns the intellectual property rights to this is the Fall 2024 Wire Strander Team 15, consisting of Luke Bottger, Cody Clifford, Karen Hamaguchi, Kendrick Langenbach, and Ted Yee, under the advising of Professor Scott Julien. The brand name and logo  were created for this group and is not a business entity. The team can be contacted via:

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j.gager@northeastern.edu

Following completion of the project, ownership of the machine has been granted to Northeastern Professor Ahmad Nourian Avval for materials research and shipped to the Additive Manufacturing Laboratory at the following address:

c/o Jon Gager

Northeastern Innovation Campus, Building 5

147 South Bedford Street

Burlington, MA, USA

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Northeastern University
College of Engineering

